

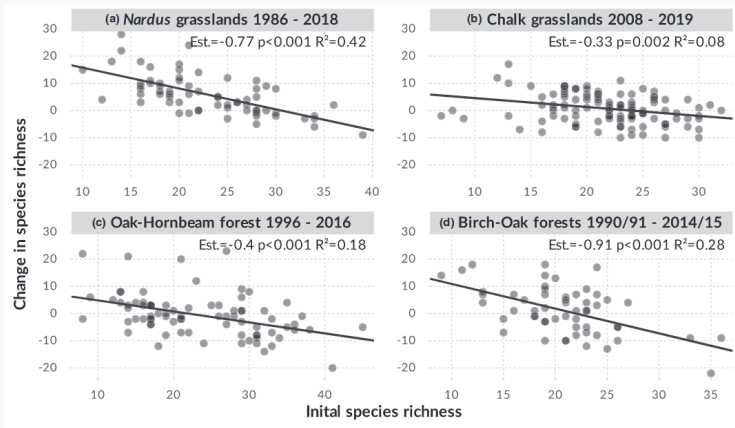
Regression to the mean

Francisco Rodríguez-Sánchez

<https://frodriguezsanchez.net>

The most biodiverse sites are losing more species

WHY??



Mazalla & Diekmann 2022

Most biodiverse sites are losing more species. Why?

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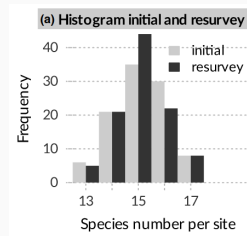
- Stronger competition
- Humans destroying most species-rich sites
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Most biodiverse sites are losing more species. Why?

- Stronger competition
- Humans destroying most species-rich sites
- Establishment of new species favoured in poor sites
- No ecological cause, but stochastic variation (**regression to the mean**)

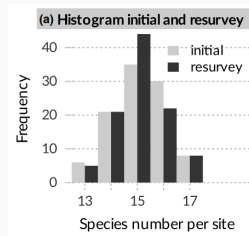
A simulation for 100 sites

- Simulate initial number of species:



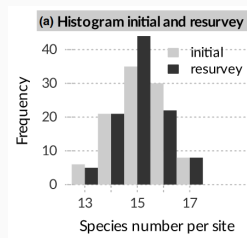
A simulation for 100 sites

- Simulate initial number of species:
 - `rnorm(n = 100, mean = 15, sd = 1)`



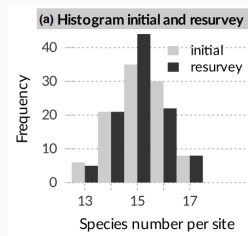
A simulation for 100 sites

- Simulate initial number of species:
 - `rnorm(n = 100, mean = 15, sd = 1)`
- Simulate number of species at resurvey:



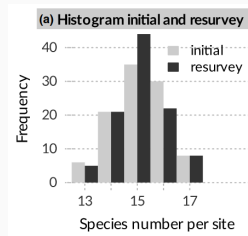
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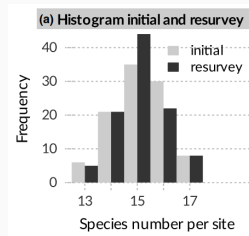
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A simulation for 100 sites

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 - `rnorm(n = 100, mean = 15, sd = 1)`
- Simulate number of species at resurvey:
 - `rnorm(n = 100, mean = 15, sd = 1)`
- **No real change at all!**
- (only stochastic variation)

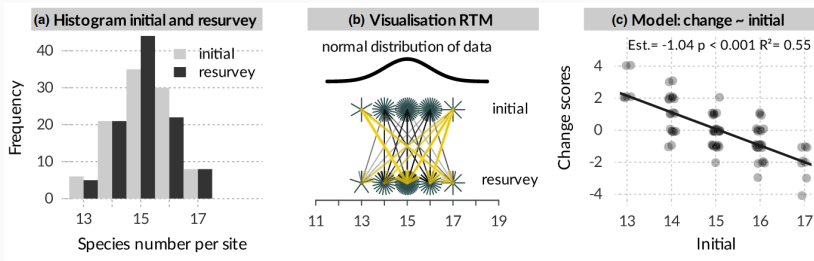


Regression to the mean

Species-rich sites lose more species

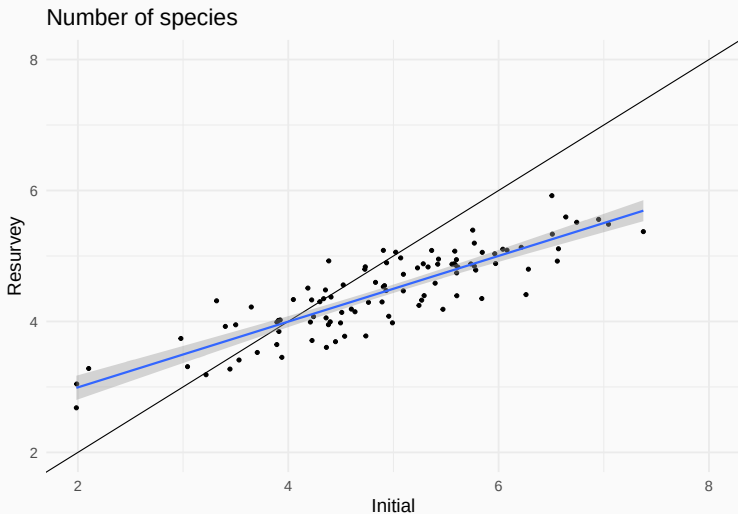
Species-poor sites gain more species

Negative trend against baseline



Mazalla & Diekmann 2022

Whenever two sets of measurements are not perfectly correlated
there will be regression towards the mean



What to do?

- Model outcome ~ baseline

What to do?

- Model outcome ~ baseline
- If modelling Change, include baseline as predictor

To learn more

- [Mazalla & Diekmann 2022](#)

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- [Mazalla & Diekmann 2022](#)
- [Kelly & Price 2005](#)